Optical Communication Systems

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***Abstract*—Optical communication systems refer to systems that utilize optical signals to establish communication between two points. It is an important area to research as it has transformed our lives by forming the basis of global -and some local- communication networks, such as, among other things, the internet, due to it being able to carry signals over very long distances. In this paper we aim to explore what properties of optical communication that gives it this capability, of what is it composed, and what are its characteristics and forms through research of papers primarily published by the IEEE and the compilation of their findings.**

Keywords—Optical communication, fiber-optic communication, free-space optical (FSO) communication, optical wireless communication (OWC), wavelength-division multiplexing (WDM).

I. Introduction

Optical communication system is a communication system where light is used as a carrier to transmit and receive data. Such communication method has been used throughout human history, even in ancient times.

Recently, however, advancements in technology have allowed the utilization of this fast medium in a massive scale for modern global communication. Given that nothing can exceed the speed of light and knowing that light is unaffected by electromagnetic interference, light would naturally be a perfect communication signal carrier. Therefore, optical communication has been an appealing system for data transmission.

Another appealing factor of optical communication systems are optical fibres, the medium which carries light signals from one point to another. Since it is made of glass, this makes it a relatively inexpensive medium, small in cross sectional area, and both strong and flexible, not to mention the infinitesimal distortion factor it adds to the data being transmitted, making it a medium that competes with coaxial cables and conventional wires in many areas of communication as a medium [1]. As a result of this, fibre optical communication systems have both a high capacity [2] and very little transmission loss.

Today, thanks to optical communication systems, we are provided with a communication medium that spans through the oceans between continents upon whom the global communication system we call “The internet” is based on. Thus, it is safe to say that optical communication systems have transformed our lives radically and affected how we all interact with each other. This gives it a high importance to be researched and developed more to enhance its capabilities. In this paper we will be discussing how optical communication systems work and what gave it the ability to transform our lives on earth.

II. Fiber-Optic Communication

The fiber-optic communication link shares similarities with other conventional communication system links. The most basic form of it consists of a transmitter, a receiver, and an optical glass or silica fiber as a transmission medium.

The transmitter converts any digital or analog signal into a discrete light pulse (pulse modulation), or an intensity based light signal (amplitude modulation) respectively [3]. For this, a light producing diode is usually used, as they emit wavelengths that are the same as the wavelength region of the attenuation for the low-loss fiber [1]. Most commonly a laser producing diode is used, as it is more efficient than a non-laser light emitting diode, such as LEDs [3].

After the signal makes its trip through the optical fiber, it is received by a receiver at the end, where it gets amplified and demodulated to return as an electrical signal that can be interpreted by the electronic equipment and be decoded.

While it is important to have fiber with high capacity, it is also important to have fast optical transmitters and receivers to keep up with its speed.

A. Fiber Cable

In the late 1960s low-loss silica fiber was proposed as a medium for optical communication transmission. By the late 1970s, research and development of this medium produced silica fiber with the loss of 0.2dB/km, the lowest theoretical limit of silica fiber. This allowed for low-loss silica to outperform any other metal-based medium that existed at the time, all in a fiber that could be as thin as a human hair [3]. This breakthrough allowed for the development of systems that work with optical communication, as interest into the potential of such systems has gone up exponentially as time went on.

Fiber optics’ transmitting principle depends upon the concept of total internal reflection. Where if a ray of light is going from one medium to another, and if the angle between it and the second medium is sufficiently low enough, instead of passing to the second medium, the ray of light will instead be completely reflected by it, and it remains in the first medium [4]. A fiber optic utilizes this property to its benefit as illustrated in fig. 1.

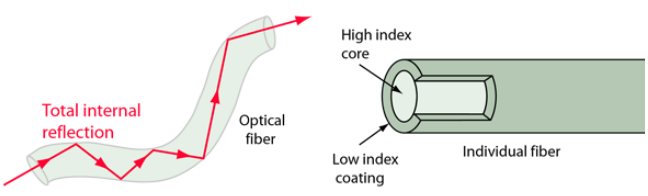

Fig. 1. The fiber optic’s topology and its total internal reflection visualized.

As also seen in fig. 1, a fiber cable consists of a core with a high refractive index in the center surrounded by coating (cladding) with a low refractive index to insure full internal reflection. The fiber cable is also surrounded by another coating that provides a protective buffer from the outside to prevent both the escaping of the signal or the entrance of noise.

B. *Optical Amplification*

Optical amplifications are required for long distance optical communication to strengthen a weakened signal and prolong the distance it can travel through the fiber without arriving heavily lost and incomprehensive. Such devices are crucial for today's global communication networks such as the internet.

Optical amplifiers came as a replacement for optical-electrical-optical repeaters (OEO), which reconvert the optical signal to an electrical signal and then regenerates it with increased signal power. The reason behind of which is the priciness of OEO and the difficulty to construct and implement them. In addition, a single optical amplifier can work with several modulations and bandwidths of signals, while EOE are limited to only one of them.

There are many different techniques of optical amplification, these include erbium-doped waveguide amplifiers, semiconductor optical amplifiers, and Raman amplifiers [5].

C. *Transmitter*

Optical transmitters are devices that receive an electrical signal and converts it to an optical one by producing light waves. As such, it is logical to use a light producing semiconductor device, as they are the most efficient and powerful ones available [6].

Laser diodes, however, are especially more effective for applications such as an optical transmitter, as it produces coherent light, unlike light emitting diodes (LEDs) that produce incoherent light. While LEDs had been considerably used despite of their disadvantage due to them being more economic-friendly because of their simplicity, that changed with the development of vertical cavity surface emitting laser (VCSEL), which combines the effectiveness of previous laser diodes and the inexpensiveness of LEDs [7], thus having laser diode technology superseding that of LEDs in terms of the application of optical transmitters.

D. *Receiver*

Receivers are devices which convert the optical signal into an electrical one, to produce the electrical signal that existed before becoming optical. Such devices utilize components called photodetectors or photosensors.

In optical communication systems, like optical transmitters, optical receivers also utilize semiconductor elements to carry out their function. However, the semiconductor is coupled with a transimpedance amplifier and a limiting amplifier to produce a properly converted electrical signal.

In optical systems that utilize coherent light signals, there can also be a local oscillator laser to help with the conversion process.

E. *Wavelength-Division Multiplexing*

Wavelength-division multiplexing (WDM) is a method which increases the bandwidth that can be carried by optical modulators and detectors [8].

In this technique, a single carrier can carry several wavelengths of light at the same time, that then can be demultiplexed at the receiver [8]. This allows for more bandwidth to be obtained and it allows for bidirectional communication in a single optical fiber to be possible [9] [10], a process called wavelength-division duplexing. Optical filters can be used for multiplexing and demultiplexing the light signals, such as: the interference filter, the prism, and the grating filter [9].

There are two configurations for WDM transmission systems, one-way and two-way. In one-way transmission, only a multiplexer and a demultiplexer are required at the transmitter and the receiver respectively. While in two-way transmission, both must be at the transmitter and the receiver [10]. This is shown in fig. 2.

The WDM technique is popular in telecommunication as it increases the capacity of the network without the need to increase the number of channels.

WDM systems can be divided to 3 types depending on their wavelength patterns:

- Normal WDM: Uses normal wavelengths 1315 nm and 1550 nm.
- Dense WDM (DWDM): Uses the C-band (1530 nm-1565 nm) transmission window but with denser channel spacing.
- Coarse WDM (CWDM): Uses 16 different transmission windows.

F. *Fiber-Optic Channel Capacity*

Channel capacity is the measure of the quality of performance of communication systems, where bandwidth and signal to nose ratio (SNR) are taken as parameters for it.

$$C = W \log_2(1 + \frac{S}{N}) \qquad (1)$$

The channel capacity can be found using (1), where W represents the bandwidth, and S/N represents signal to noise power ratio [2][11].

In conventional communication systems, this method of finding the channel capacity for linear systems is acceptable.

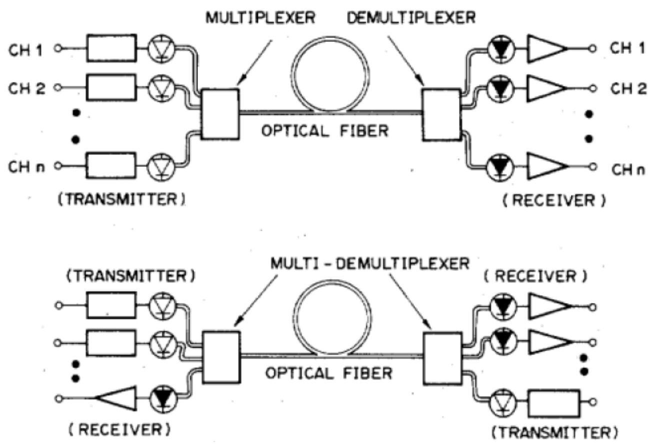

Fig. 2. Modules showing a one-way WDM system (first) and a two-way WDM system (second).

However, when it comes to modern optical fiber systems, they operate on non-linear regions and thus (1) is insufficient for finding the channel capacity they possess [2][11].

The required high amplitude of the pulse signals and the increase in the number of wavelength division multiplexing (WDM) channels present in the system lead to an increase in the electric field intensity in the fiber and thus substantially brings the effect of the Kerr non-linearity of the refractive index in the system to a level that cannot be ignored [2][10].

$$n = n_0 + n_2 I \tag{2}$$

The Kerr nonlinearity effect can be calculated using (2), where n_0 represents the linear refractive index, n_2 the second order nonlinear refractive index, and I the wave intensity [12].

Another factor that significantly affects channel capacity in fiber optic communication is the presence of filtering effects. Those effects can come in the form of two sources, all-pass and bandpass filters. The first one originates from the dispersive nature of optical fibers, which is chromatic dispersion. While the other comes from the presence of optical bandpass filters [2].

G. Comaprison with Wired Communicaion Systems

As discussed above, fiber optic communication systems show superiority to wired communication in many aspects. These include:

- Lower material cost for fibers, so long as their quantities are not large [3].
- Lower cost of transmitters and receivers.
- Capability to carry electrical power as well as signals if desired.
- simplicity of operating transducers in linear mode.
- Immunity to electromagnetic interference, which eliminates distortion [3].
- High electrical resistance, making it safe to use near high-voltage equipment.
- Lighter weight and more flexibility [1] [3].
- No sparks, eliminating danger especially in flammable or explosive gas environments [3].
- Not electromagnetically radiating, and difficult to tap without disrupting the signal, which considerably increases security [3].
- Much smaller cable size—important where pathway is limited [3].
- Resistance to corrosion due to non-metallic transmission medium.

Having listed all of this, it is easy to see why optical communication systems have simplified and significantly enhanced communication both on a local and global levels.

III. Free-Space Optical Communication

Free-Space optical communication (FSO) system, also known as optical wireless communication, utilizes free-space (such as air or vacuum), rather than optical fibers, to transmit optical signals. Such system can be used to transmit data so long as there is no nontransparent object to block the signal midway through transmission. Transmission can be achieved in two ways, either in a diffused one where the light is emitted incoherently in the area [13], or in a focused way where a laser is emitted directly to the receiver [14].

While the nontransparent blockage hinderance might be considered a disadvantage, it still means that, unlike radio signals, optical signals can still penetrate through transparent objects such as glass, and additionally this property can make for a secure, untappable channel that is confined in the walled space it exists in [13], so it can act as an advantage in particular situations.

FSO technology has always been limited in the distance it can travel through and the amount of data rate it can possess. Although advancements in laser technology helped it, it is still suffering from problems which limits the use applications it can have. Not to mention that conditions such as atmospheric ones can have an effect in the performance of the system [15]. This system can only be employed when the transmitter and the receiver are not beyond several kilometers in distance away from each other [14], as other than that the signal would become incomprehensible and lost.

A. Applications

While it is indeed limited in use, free-space optical communication system has still found usefulness in many areas, and indeed demonstrated effectiveness in accomplishing its functions. This can be seen, for example, in the first ever wireless phone, the photophone, invented by the same person who invented the wired phone, Alexander Graham Bell. As its name suggests, it utilizes optical communication to establish voice transmission, specifically FSO communication, and Bell considered it to be the greatest of his inventions [14].

Moreover, an area that showed great interest towards FSO communication systems is the military, having always endeavored to develop the technology to implement in its covert operations [16]. Another field that found use in FSO communication is the space sector, where it is in consideration and development to be used for inter-satellite and deep-space links. In addition, the telecommunication industry has recently been considering it a solution for the last-mile problem to link between the commercial user and the fiber-optic infrastructure [14].

While it has historically been known to be limited in data rate, recent developments show prospects of it competing with fiber-optic communication systems [14], so it is important to note that it is not a system fully devoid of potential, but rather one that lacked research and development ever since the rise of fiber-optic technology shifted the focus and resources of researchers towards itself away from FSO.

IV. Conclusion

In this paper we have endeavored to explain the mechanisms of optical communication systems, their categories, the devices involved in them, and their importance in our daily lives. We have explained how the transmitter and the receiver of such systems are, for the most part, simple light producing and photo detecting semiconductors respectively. The transmitter and the receiver are connected by fiber-optics that can transmit light between them by utilizing the total internal reflection property which maintains signal strength through long distances, aided by optical amplifiers. We have also shown the great channel capacity capability of fiber optics due to signal maintenance and not being affected by electromagnetic interference (EMI), despite nonlinear effects such as the Kerr and filtering ones making an appearance exclusively in fiber-optics. The WDM technique of transmitting signals is also mentioned, which greatly aids in increasing the channel capacity of the fiber as it allows several signals to travel in the same channel at the same time by being multiplexed, and how it allows for bidirectional connection.

We have also briefly explored the other form of optical communication, which is FSO communication. We have demonstrated that while fiber-optic communications are a more popular and effective from of optical communication, FSO communication is merely a neglected area of research that still holds a lot of potential and is applied by several important fields, even if noncommercial.

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